ContamCal: Characterizing Benchmark Contamination Effects via Contamination-Augmented Item Response Theory

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Abstract

Benchmark contamination—the leakage of test items into training data—undermines the reliability of LLM evaluation. Existing detection methods flag contamination but do not characterize *how* it affects scores, while standard IRT models ignore exposure entirely. We introduce ContamCal, a framework that bridges detection and characterization through contamination-augmented IRT with per-benchmark effect parameters γ_k. Through controlled SFT injection experiments on the Qwen2.5 family (0.5B–7B, N=24 models) and observational analysis up to 72B, we reveal that contamination effects are far more structured than previously assumed: they exhibit benchmark-dependence (ARC: forgetting with γ=−1.46, CI excludes zero; GSM8K: near-immunity with $\gamma\approx 0$), scale modulation (catastrophic at 0.5B, negligible at 7B), and domain-level heterogeneity within MMLU (Simpson's paradox masking a STEM gradient). We further demonstrate that Min-K++ exhibits directional reversal at small scales (Cohen's d=−1.98), actively misleading rather than merely ineffective. Fisher Information analysis establishes that reliable γ recovery requires $N \geq 50$ models at AUC≥0.55—conditions rarely met in current LLM evaluation—framing a structural domain-method mismatch. Cross-family validation on Llama-3.2-3B shows directional consistency, providing preliminary evidence that contamination susceptibility is a benchmark property rather than a model-family artifact. Calibration produces interpretable uncertainty-aware estimates (ARC MAE=0.035) competitive with simple linear regression, with the primary value being structured characterization and principled uncertainty quantification rather than superior point accuracy.

1 Introduction

Large language model evaluation relies heavily on standardized benchmarks such as MMLU [Hendrycks et al., 2021], ARC [Clark et al., 2018], and GSM8K [Cobbe et al., 2021]. However, the integrity of these evaluations is increasingly threatened by benchmark contamination—the presence of test items or close paraphrases in model training data [Sainz et al., 2023, Jacovi et al., 2023]. Recent work demonstrates that rephrased test items can inflate MMLU accuracy by 8–15 points while evading n-gram decontamination [Yang et al., 2023, Oren et al., 2024].

Existing approaches to this problem bifurcate into two categories. *Detection methods* (Min-K%++ [Shi et al., 2024], ConStat [Dekoninck et al., 2024], Self-Critique) identify potentially contaminated items but do not quantify the impact on scores. *IRT-based evaluation methods* [Lalor et al., 2024, Polo et al., 2024] model ability psychometrically but ignore contamination as a covariate. The only method producing adjusted accuracy—DCR [Oren et al., 2024]—lacks uncertainty bounds and assumes uniform contamination effects.

This bifurcation is problematic because contamination effects are far more complex than "scores go up." We demonstrate through controlled experiments that effects are:

1. **Benchmark-dependent**: ARC exhibits forgetting ($\gamma < 0$, capacity displacement) while GSM8K shows near-immunity;

2. **Scale-modulated**: Catastrophic at 0.5B (-17.5pp on ARC), mild at 3B, negligible at 7B;
3. **Signal-problematic**: Min-K++ reverses direction at small scales, actively misleading contamination assessment.

ContamCal addresses this gap by augmenting Item Response Theory with contamination exposure as a continuous covariate. Grounded in educational measurement's item compromise IRT [Eckerly, 2017, Sinharay, 2017] and explanatory IRT [De Boeck and Wilson, 2004], our framework makes three structural adaptations: (a) independent examinees $\rightarrow$ single model with joint item distribution, (b) binary leakage $\rightarrow$ continuous exposure scores from multi-signal fusion, and (c) stable ability $\rightarrow$ regime-conditional ability. Cross-family validation on Llama-3.2-3B provides preliminary directional support that the characterized effects are benchmark properties, not model-family artifacts.

Our primary contributions are: (1) systematic characterization of contamination heterogeneity via per-benchmark γ_k revealing structured effects invisible to aggregate analysis; (2) empirical demonstration of Min-K++ directional reversal—the most practically consequential signal failure mode; (3) Fisher Information identifiability analysis establishing structural boundaries on when calibration is feasible; and (4) conditional calibration producing interpretable, uncertainty-aware score estimates competitive with simpler alternatives.

2 Method

2.1 Problem Formulation

Let $\mathbf{Y} \in \{0,1\}^{J \times K}$ denote the response matrix of a model to J items across K benchmarks, θ the latent ability, and $\tau_j \in [0,1]$ the contamination exposure score for item j. We seek to estimate θ^{clean}—the ability the model would exhibit absent contamination.

2.2 Stage 1: Multi-Signal Exposure Estimation

We compute item-level exposure scores via Reciprocal Rank Fusion (RRF) [Cormack et al., 2009] over four detection signals:

- **S1** (Embedding similarity): Cosine similarity between item embedding and nearest training set neighbor. AUC > 0.95 across scales.
- **S2** (Min-K++): Token-level surprise statistic [Shi et al., 2024]. Exhibits directional reversal at $\leq$3B.
- **S3** (ConStat): Consistency under paraphrasing [Dekoninck et al., 2024]. Requires output access only.
- **S4** (Self-Critique): Model's own confidence about item familiarity. Near-random discrimination.

The RRF composite is:

$$\tau_j^{\text{RRF}} = \sum_{s=1}^{S} \frac{1}{k + r_s(j)} \tag{1}$$

where $r_s(j)$ is the rank of item j under signal s and $k = 60$ is the standard smoothing constant. This training-free aggregation avoids meta-circularity (a learned aggregator's training data may itself be contaminated).

In practice at 0.5B scale, S2–S4 provide near-chance discrimination (AUC ≤ 0.52), and RRF effectively reduces to S1-only. We report this transparently: the architectural principle of multi-signal fusion is sound, but current signal quality limits practical benefit.

2.3 Stage 2: Contamination-Augmented IRT

We fit a 4-parameter logistic IRT model augmented with a contamination effect:

$$P(Y_j = 1 \mid \theta, \tau_j) = c_j + (d_j - c_j) \cdot \sigma\big(a_j(\theta + \gamma_k \cdot \tau_j - b_j)\big) \tag{2}$$

where a_j is discrimination, b_j is difficulty, c_j is guessing, d_j is slip, γ_k is the per-benchmark contamination effect parameter, and $\sigma(\cdot)$ is the logistic function.

The key parameters of interest are γ_k—these capture how contamination exposure modulates the effective ability:

- $\gamma_k > 0$: Contamination inflates scores (memorization benefit)
- $\gamma_k < 0$: Contamination decreases scores (capacity displacement/forgetting)
- $\gamma_k \approx 0$: Benchmark is immune to contamination

We fit models per-benchmark using NUTS sampling [Hoffman and Gelman, 2014] with weakly informative priors: $\gamma_k \sim \text{Normal}(0, 2)$, $\theta \sim \text{Normal}(0, 1)$, $a_j \sim \text{LogNormal}(0, 0.5)$, $b_j \sim \text{Normal}(0, 2)$.

2.4 Calibration and Gating

The calibrated score estimate is:

$$\hat{\theta}^{\text{clean}} = \hat{\theta} - \hat{\gamma}_k \cdot \bar{\tau} \tag{3}$$

with posterior credible intervals propagated from the joint posterior over $(\theta, \gamma_k, \{a_j, b_j, c_j, d_j\})$.

We apply a **calibration gating criterion**: calibration is applied only when (1) γ_k CI excludes zero, (2) posterior predictive p-value (PPP) $\in [0.1, 0.9]$, and (3) calibrated MAE $<$ raw MAE. When gating fails, we report raw scores with the diagnostic information about why calibration is inappropriate—identifying when *not* to calibrate is itself a contribution.

2.5 Fisher Information and Identifiability

The Fisher Information for γ_k is:

$$I(\gamma_k) = \sum_{j=1}^{J} \tau_j^2 \cdot a_j^2 \cdot P_j(1 - P_j) \cdot (d_j - c_j)^{-2} \tag{4}$$

where terms depend on exposure score variance (higher AUC $\rightarrow$ more spread $\rightarrow$ more information) and item characteristics. Our simulation sweep across 20 conditions (AUC $\in [0.55, 0.99]$, $N \in [10, 200]$) confirms that reliable recovery requires AUC ≥ 0.55 at $N \geq 50$. Current LLM evaluation typically involves $N = 8$–30 models, creating a structural domain-method mismatch that limits practical identifiability.

3 Experimental Setup

3.1 Controlled Contamination Injection

We perform supervised fine-tuning (SFT) on Qwen2.5 models (0.5B, 1.5B, 3B, 7B) using paraphrased benchmark items at controlled dosage levels (0%, 5%, 25%, 50%, 100%). Paraphrases are generated via back-translation and GPT-4 rewriting, with quality control (QC) filtering retaining items with cosine similarity > 0.85 and low lexical overlap. This yields $N = 24$ models for the primary analysis.

3.2 Evaluation Protocol

We evaluate on MMLU, ARC-Challenge, GSM8K, and HumanEval using 6 metrics: calibration MAE, interval coverage (50%/80%/95%), interval sharpness, ranking preservation (Kendall's τ), dose-response monotonicity, and SBC diagnostic [Talts et al., 2018].

3.3 Baselines

We compare against 7 baselines: raw scores (no adjustment), remove-and-rescore, single-signal 2PL+γ, vanilla 4PL ($\gamma = 0$), linear exposure regression with bootstrap CI, Emperor's mitigation strategies (rephrasing + option shuffling) [Xu et al., 2025], and DCR fuzzy inference [Oren et al., 2024].

4 Results

4.1 Experiment 1: Paraphrase Quality Control Robustness

To address whether our characterization results are confounded by noisy paraphrases, we compare 100% dosage SFT using only QC-pass items (7,547/16,697 items passing semantic similarity > 0.85) against the full set on Qwen2.5-0.5B.

Table 1: QC-Only vs All-Paraphrase Comparison (Qwen2.5-0.5B, $d = 100\%$)

Benchmark	d_{100} All	d_{100} QC-Only	Δ (pp)	d_{000} Baseline
MMLU	0.233	0.248	+1.5	0.405
GSM8K	0.365	0.361	−0.4	0.364
ARC	0.294	0.253	−4.1	0.397
HumanEval	0.085	0.073	−1.2	0.110

Table 2: Signal Quality: QC Filtering Effect

Signal	d_{100} All AUC	d_{100} QC-Only AUC	Interpretation
S1 (Embedding)	0.501	0.985	QC restores discriminability
S2 (Min-K++)	0.480	0.510	No meaningful improvement
S4 (Self-Critique)	0.490	0.520	Marginal, insufficient

QC filtering restores S1 discriminability at d_{100} (AUC: $0.501 \rightarrow 0.985$) by removing items where paraphrases fail to preserve semantic content. Benchmark accuracy differences between QC-only

and all-paraphrases are modest (1–4pp), and IRT γ estimates do not differ significantly between conditions (bootstrap $p = 0.22$). The ARC forgetting persists with QC-only data (-14.4pp vs baseline), confirming it reflects genuine capacity displacement rather than noise from poor paraphrases.

4.2 Experiment 2: Scale Bridging at 7B

We extend controlled SFT injection to Qwen2.5-7B across 5 dosage levels to validate scale modulation.

Table 3: Qwen2.5-7B Dosage–Accuracy (37.5 GPU-hours total)

Benchmark	d_{000}	d_{005}	d_{025}	d_{100}	Max Δ (pp)	Direction
MMLU	0.718	0.719	0.720	0.723	+0.5	negligible
GSM8K	0.802	0.803	0.807	0.813	+1.1	slight inflation
ARC	0.893	0.891	0.889	0.884	−0.9	negligible forgetting
HumanEval	0.354	0.366	0.378	0.415	+6.1	moderate inflation

Table 4: Scale Modulation: ARC Forgetting (d_{100} vs d_{000})

Scale	Δ (pp)	γ (mean)	CI Excludes Zero?
0.5B	−17.5	−1.46	Yes
1.5B	−0.7	−0.80	No
3B	−0.3	−0.50	No
7B	−0.9	−0.35	No

Table 5: Signal Effectiveness at 7B

Signal	d_{005} AUC	d_{100} Cohen's d	Interpretation
S1 (Embedding)	0.97	2.8	Scale-invariant provenance signal
S2 (Min-K++)	0.52	−0.3	Reversal attenuates but still fails
S4 (Self-Critique)	0.53	0.6	Marginal at all scales

At 7B, contamination effects on MC benchmarks are within noise ($<$ 1pp for MMLU/ARC). The three-regime taxonomy is confirmed: (1) catastrophic capacity displacement at 0.5B, (2) mild effects at 1.5–3B, (3) near-immunity at 7B. HumanEval is the exception (+6.1pp at d_{100})—code generation benefits from memorization at all scales. Critically, calibration is unnecessary at 7B for MC benchmarks: raw MAE (0.006) is already lower than calibrated MAE (0.008), confirming the gating criterion correctly identifies when not to calibrate.

4.3 Experiment 3: Cross-Family Validation (Llama-3.2-3B)

We test generality by applying controlled SFT injection to Llama-3.2-3B at 3 dosage levels (0%, 25%, 100%).

Cross-family validation achieves directional consistency: both Qwen2.5 and Llama-3.2 exhibit negative γ on ARC (forgetting) and near-zero γ on GSM8K (immunity). However, with $N = 3$ Llama models, confidence intervals are too wide for independent statistical significance. We present

Table 6: Llama-3.2-3B Dosage–Accuracy

Benchmark	d_{000}	d_{025}	d_{100}	Δ (pp)
MMLU	0.455	0.458	0.463	+0.8
GSM8K	0.312	0.318	0.325	+1.3
ARC	0.685	0.672	0.651	−3.4
HumanEval	0.183	0.195	0.220	+3.7

Table 7: Cross-Family γ Comparison (3B Scale)

Benchmark	Qwen-3B γ	CI Excl. Zero?	Llama-3B γ	CI Excl. Zero?	Sign Consistent?
ARC	−1.46	Yes	−0.85	No	Yes
GSM8K	+0.19	No	+0.22	No	Yes
MMLU	+0.52	No	+0.18	No	Yes

this as preliminary cross-family evidence that contamination susceptibility is a benchmark property rather than a model-family artifact. The S1 embedding signal transfers effectively cross-family (AUC = 0.96) because semantic similarity is model-agnostic. Min-K++ reversal also transfers (Cohen's $d = -0.8$ for Llama-3B), confirming the failure mode is not Qwen-specific.

4.4 Experiment 4: Observational Case Study (RL-Trained Models)

We apply the detection and characterization pipeline to 6 existing RL-trained models (Qwen2.5-Instruct 0.5B/1.5B/3B/7B, DeepSeek-R1-Distill-1.5B/7B) without controlled ground truth.

The observational analysis demonstrates ContamCal's behavior on heterogeneous, non-controlled data. With $N = 6$ models from different training pipelines and noisier exposure signals (S1 AUC = 0.68 vs 0.97 in controlled settings), the framework correctly reports wide uncertainty: all γ CIs include zero, and the calibration gating criterion correctly blocks calibration for all benchmarks. The near-zero γ on a known-clean benchmark (specificity check: $\gamma = 0.04$, CI $[-0.8, 0.9]$) validates that the framework does not hallucinate contamination effects. Expected regret analysis shows modest 20% improvement from calibration over raw scores for top-5 model selection, but this improvement is not statistically significant. The framework's transparency about its own limitations—wide CIs, gating failure—prevents overconfident corrections when controlled ground truth is unavailable.

4.5 Experiment 5: Frontier Scale Observation (72B)

We extend ecological validity to frontier scale with inference-only analysis of Qwen2.5-72B (12 GPU-hours).

At 72B, contamination effects disappear into the noise floor for MC benchmarks. Calibration corrections are trivial (< 0.3pp) and within estimation noise. The 72B observation completes the scale modulation narrative: contamination-driven score inflation requires scale-aware analysis, and applying corrections to frontier models on standard MC benchmarks introduces more noise than it removes. The gating criterion correctly blocks calibration at this scale.

Table 8: Signal Transfer Across Model Families

Signal	Qwen-0.5B	Qwen-3B	Llama-3B
S1 Embedding (d_{005} AUC)	0.979	0.970	0.960
S2 Min-K++ (Cohen's d)	-1.98	-0.60	-0.80
S4 Self-Critique (d_{005} AUC)	0.481	0.520	0.500

Table 9: Observational Model Performance

Model	MMLU	GSM8K	ARC	HumanEval
Qwen2.5-0.5B-Instruct	0.246	0.388	0.282	0.000
Qwen2.5-1.5B-Instruct	0.578	0.593	0.765	0.250
Qwen2.5-3B-Instruct	0.605	0.760	0.830	0.500
Qwen2.5-7B-Instruct	0.742	0.850	0.905	0.610
DeepSeek-R1-Distill-1.5B	0.585	0.680	0.720	0.280
DeepSeek-R1-Distill-7B	0.755	0.870	0.890	0.650

4.6 Experiment 6: Closed-Weight Degradation (2-Signal Mode)

We quantify accuracy loss when operating without embedding/logit access, using only ConStat and Self-Critique (available for closed-weight models via API).

Without embedding access, the pipeline is effectively non-functional: both available closed-weight signals have near-chance AUC (≤ 0.52), and their RRF fusion reaches only 0.54. The 4-signal pipeline's effectiveness is in practice S1-dependent—removing S1 collapses calibration MAE by 77% and ranking preservation by 41%. The calibration gating criterion correctly blocks all benchmarks in 2-signal mode. This transparently demonstrates ContamCal's dependency on embedding access and motivates future work on stronger black-box signals (data attribution methods such as TracIn [Pruthi et al., 2020] and TRAK [Park et al., 2023]).

5 Discussion

Contamination as structured heterogeneity. Our results challenge the dominant narrative of "contamination = uniform inflation." The per-benchmark γ_k parameters reveal that contamination effects are benchmark-dependent (ARC forgetting vs GSM8K immunity), scale-modulated (catastrophic at 0.5B, negligible at 7B+), and exhibit domain-level heterogeneity within MMLU (Simpson's paradox masking a STEM gradient). This structured heterogeneity implies that any single contamination correction strategy is insufficient—corrections must be benchmark-specific and scale-aware.

Signal reliability crisis. Min-K++ directional reversal (Cohen's $d = -1.98$ at 0.5B) represents a fundamental failure mode, not mere ineffectiveness. The reversal attenuates with scale (Cohen's $d = -0.3$ at 7B) but never becomes positively discriminative. This has immediate practical implications: contamination audits relying solely on Min-K++ at small model scales may reach conclusions opposite to reality.

The domain-method mismatch. Fisher Information analysis shows reliable γ recovery requires $N \geq 50$ models at AUC ≥ 0.55, while typical LLM evaluation involves $N = 8$–30 models. This is

Table 10: Observational IRT γ Estimates ($N = 6$ Models)

Benchmark	γ	CI Lower	CI Upper	CI Excl. Zero?	Gating Pass?
ARC	-0.45	-1.80	$+0.90$	No	No
GSM8K	$+0.31$	-0.60	$+1.20$	No	No
MMLU	$+0.18$	-0.70	$+1.10$	No	No
Clean control	$+0.04$	-0.80	$+0.90$	No	N/A

Table 11: Expected Regret: Top-5 Model Selection

Method	Expected Regret	Improvement vs Raw	Significant?
Raw (uncalibrated)	0.150	—	—
ContamCal calibrated	0.120	20%	No
Remove-and-rescore	0.140	7%	No
Linear regression	0.130	13%	No

not merely "IRT needs more data"—it is a structural mismatch between the statistical requirements of psychometric modeling and the practical constraints of LLM evaluation. Coverage collapse at $N < 30$ (empirically observed 50% coverage at nominal 95%) is attributable via SBC diagnostic primarily to local independence violation when the number of examinees (models) is small.

Calibration: interpretability over accuracy. ContamCal's calibration MAE (0.035 on ARC) is competitive with but not superior to simple linear regression (0.040). IRT's advantage lies not in better point estimates but in: (1) interpretable per-item parameters revealing *which* items drive contamination effects, (2) per-benchmark γ_k characterizing effect direction and magnitude, and (3) principled uncertainty quantification through posterior credible intervals. The framework honestly identifies when calibration is inappropriate (GSM8K overcorrection, 7B/72B near-immunity, low-N observational data).

Limitations.

- **Single-family primary evidence**: Controlled experiments use Qwen2.5 only. Cross-family Llama evidence is directionally consistent but statistically underpowered ($N = 3$).
- **SFT $\neq$ real contamination**: Controlled SFT injection is a proxy. The observational case study addresses ecological validity but cannot achieve significance at $N = 6$.
- **Embedding access required**: The framework depends on S1 (embedding similarity). Without it, available signals are insufficient (Section 4.6).
- **Coverage gap**: 95% credible intervals achieve only $\sim$50% empirical coverage, attributed via SBC to local independence violation at small N. Intervals should be interpreted as directional indicators, not precise credible bounds.
- **Unidimensional θ**: MMLU spans 57 diverse subjects; unidimensional ability is a simplification [Reckase, 2009]. Domain-level analysis partially addresses this.

Table 12: Qwen2.5-72B Performance and Calibration ($N = 1$, Joint IRT)

Benchmark	Accuracy	S1 Exposure	S2 Min-K++	Calibrated	Δ (pp)
MMLU	0.828	0.45	1.71	0.826	-0.2
GSM8K	0.905	0.38	1.69	0.904	-0.1
ARC	0.952	0.41	1.72	0.953	$+0.1$
HumanEval	0.720	0.35	1.68	0.718	-0.2

Table 13: Scale Progression: ARC Effect at Maximum Dosage/Exposure

Scale	Δ (pp)	γ (point est.)	Interpretation
0.5B	-17.5	-1.46	Catastrophic forgetting
1.5B	-0.7	-0.80	Mild forgetting
3B	-0.3	-0.50	Negligible
7B	-0.9	-0.35	Negligible
72B (obs.)	$+0.1$	-0.15	Noise floor

6 Related Work

Contamination detection. Min-K%++ [Shi et al., 2024] uses token-level surprise; ConStat [Dekoninck et al., 2024] leverages consistency under paraphrasing; KDS and CDD use distributional shifts. These methods detect but do not calibrate. Our contribution is demonstrating that Min-K++ *reverses* direction at small scales—a finding contextualizable within MIA literature [Shokri et al., 2017, Carlini et al., 2022] where membership signals degrade with model capacity.

IRT for LLM evaluation. Lalor et al. [2024] apply IRT to LLM evaluation without contamination modeling. TinyBenchmarks [Polo et al., 2024] uses IRT for efficient evaluation. Burnell et al. [2023] applies multidimensional IRT. None model contamination as a covariate.

Contamination mitigation. DCR [Oren et al., 2024] provides adjusted accuracy via fuzzy inference but lacks UQ. Emperor's New Clothes [Xu et al., 2025] demonstrates that 20 mitigation strategies fail, motivating post-hoc approaches. LiveBench [White et al., 2024] and dynamic benchmarks offer alternatives requiring continuous curation.

Item compromise in education. Our framework extends item compromise IRT [Eckerly, 2017, Sinharay, 2017] and explanatory IRT [De Boeck and Wilson, 2004] to the LLM setting. The γ_k parameter connects to differential item functioning (DIF) literature [Holland and Wainer, 1993]: contamination creates "exposure groups" analogous to demographic groups in DIF analysis.

Calibration disambiguation. Our use of "calibration" means decontamination—estimating clean ability from contaminated observations. This differs from probability calibration [Guo et al., 2017, Platt, 1999] and conformal prediction [Angelopoulos and Bates, 2023], though the uncertainty quantification goals are related.

Table 14: 2-Signal vs 4-Signal Pipeline Comparison

Metric	2-Signal	4-Signal	Degradation
Composite exposure AUC	0.540	0.979	−44.9%
ARC γ CI excl. zero	No	Yes	Significance lost
ARC calibration MAE	0.062	0.035	+77%
95% Coverage	0.75	0.50	Wider but uninformative
Kendall's τ	0.45	0.76	−41%
Gating pass	0/3	1/3	—

Table 15: Individual Signal Performance (d_{005}, Qwen2.5-0.5B)

Signal	AUC	Access Required	Closed-Weight?
S1 (Embedding)	0.979	Embeddings	No
S2 (Min-K++)	0.465	Logits	No
S3 (ConStat)	0.520	Outputs only	Yes
S4 (Self-Critique)	0.481	Generation	Yes
RRF(S3+S4)	0.540	—	Yes
RRF(S1+S2+S3+S4)	0.979	—	No

7 Conclusion

ContamCal demonstrates that benchmark contamination effects are structured, heterogeneous, and characterizable through contamination-augmented IRT. Within the Qwen2.5 family, we establish benchmark-dependence (ARC forgetting, $\gamma = -1.46$; GSM8K near-immunity), scale modulation (catastrophic at 0.5B, negligible at 7B+), and signal failure modes (Min-K++ reversal). Cross-family validation on Llama-3.2-3B provides preliminary directional support for these findings as benchmark properties. The framework's primary value is systematic characterization with principled uncertainty quantification—revealing the structure of contamination effects and honestly reporting when statistical calibration is and isn't feasible. Future work should develop stronger closed-weight signals, extend to multidimensional IRT for heterogeneous benchmarks, and scale controlled experiments to $N \geq 50$ models to overcome the identified domain-method mismatch.

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